H

Roll No. .....

TCE–604

B. TECH. (CE) (SIXTH SEMESTER)

MID SEMESTER

EXAMINATION, March, 2024

CONSTRUCTION COSTING AND QUANTITY SURVEYING

Time : 1½ Hours

Maximum Marks : 50

Note : (i) Answer all the questions by choosing any *one* of the sub-questions.

(ii) Each sub-question carries 10 marks.

1. (a) What are the various types of building estimate ? (CO1)

OR

(b) Describe the various types of drawings and data required for preparation of estimating a project. (CO1)

P. T. O.

Scanned with OKEN Scanner

2. (a) What is meant by : (CO1)

(i) Bill of quantities

(ii) Annual maintenance and repair estimate

(iii) Floor area

(iv) Capital cost

OR

(b) The plinth area of an apartment is 500 sqm. Determine the total cost of building from the following data : (CO1)

(a) Rate of construction = ₹ 1230 per m^3

(b) The height of apartment = 16.23 m

(c) Water supply, Sanitary and Electrical installation each at 6% of building cost.

(d) Architectural appearance @ 1% of building cost.

(e) Unforeseen item @ 2% of building cost

(f) P.S. and contingencies @ 4% of building

3. (a) Explain the individual wall method and center line method. (CO2)

OR

(b) In Fig. 1, calculate the total quantities of the following items by using individual wall method : (CO2)

(i) Earthwork in excavation in foundation.

(ii) Brick work in foundation and plinth.

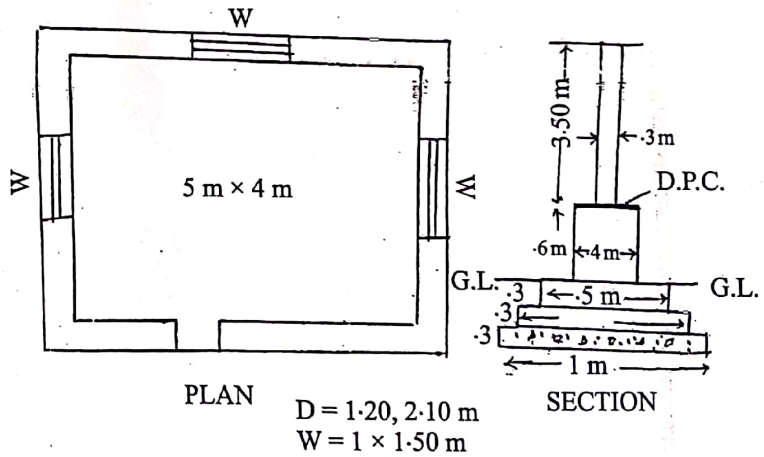
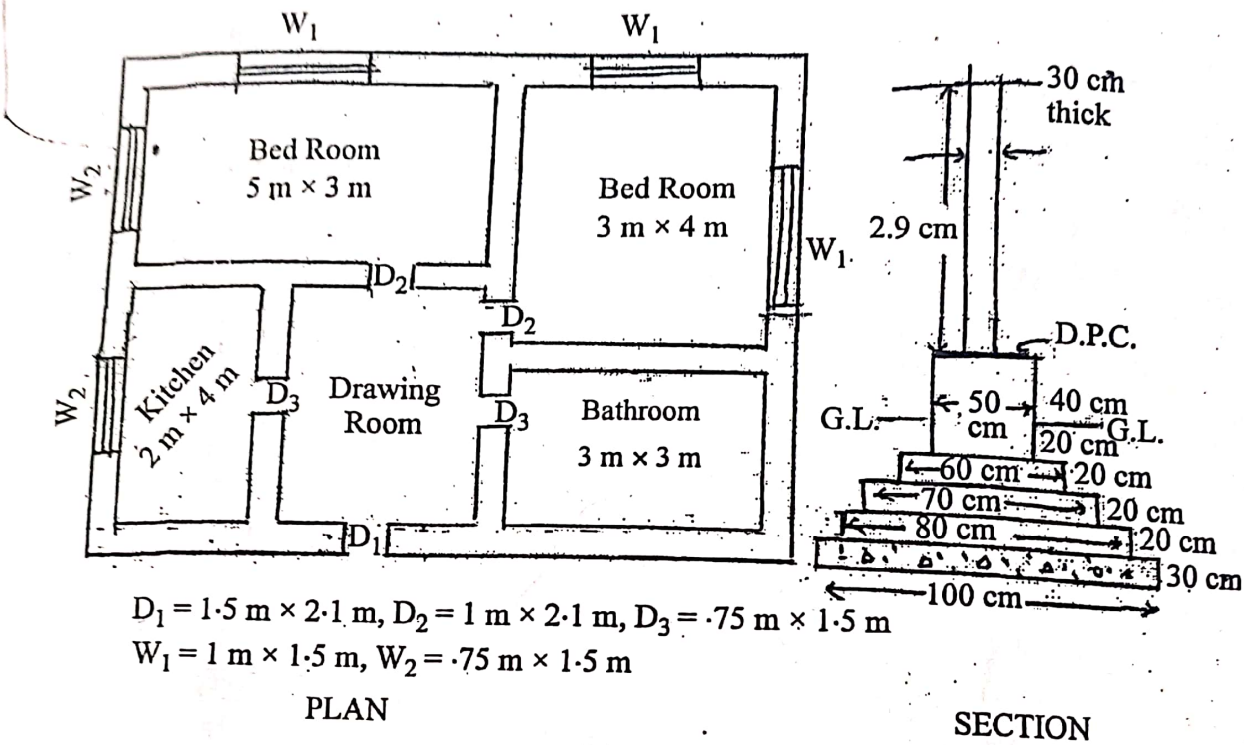


Fig. 1

Scanned with OKEN Scanner

4. (a) In Fig. 2 calculate the total quantity of Earthwork in excavation in foundation by using Individual wall method : (CO2)



$D_1 = 1{\cdot}5$ m × $2{\cdot}1$ m, $D_2 = 1$ m × $2{\cdot}1$ m, $D_3 = {\cdot}75$ m × $1{\cdot}5$ m
$W_1 = 1$ m × $1{\cdot}5$ m, $W_2 = {\cdot}75$ m × $1{\cdot}5$ m

Fig. 2

OR

(b) In Fig. 2, calculate the total quantity of lime concrete in foundation by using centre line method. (CO2)

5. (a) In Fig. 2, calculate the total quantity of 1st class brick work in foundation by using centre line method. (CO2)

OR

(b) In Fig. 2, calculate the total quantity of 1st class brick work in superstructure by using individual wall method. (CO2)

Scanned with OKEN Scanner